Local Income Taxes and Residential Sorting:

Evidence from LEHD Origin-Destination Data

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Abstract

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1 Introduction

2 Institutional Background

Local income taxes are one of the few fiscal instruments available to municipalities in Ohio and Pennsylvania. Unlike property taxes, which are paid by property owners and capitalized into home values, local income taxes fall directly on workers and vary across municipal boundaries within the same metropolitan area. This creates sharp spatial discontinuities in tax burden that can influence where people choose to live and work.

2.1 Ohio Municipal Income Tax

Ohio authorizes municipalities (cities and villages) to levy a municipal income tax on earned income under Ohio Revised Code Chapter 718. Unincorporated townships cannot levy the tax, creating a clean extensive-margin contrast between taxed and untaxed jurisdictions within the same county. As of 2019, approximately 685 municipalities levy the tax at rates ranging from 0.5% to 3.0%, with 1.0% and 1.5% being the most common. Our panel (2002–2019) contains 302 rate increases and 45 rate decreases.

Ohio operates a credit system: residents who work in a municipality with a higher tax rate than their residence receive a credit against their home municipality's tax for taxes paid to the work municipality. However, this credit system has historically been administered at the municipal level without centralized enforcement, and the credit is often partial (capped at the home municipality's rate or a fraction thereof). This means that for Ohio workers, the interaction between home and work municipality rates determines the effective tax burden, but imperfect crediting can leave workers paying more than the statutory maximum of either rate alone.

2.2 Pennsylvania Earned Income Tax

Pennsylvania's Earned Income Tax (EIT) is mandatory for all 2,500+ municipalities under the Local Tax Enabling Act (Act 511 of 1965). The EIT applies to all earned income with no minimum threshold; retirees are exempt. The standard cap is 1%, typically split 0.5% municipal and 0.5% school district. Home Rule municipalities may exceed this cap, with Philadelphia operating under a separate Sterling Act authority at approximately 3.8%. The EIT is the primary revenue lever for most Pennsylvania municipalities: property tax millage caps are binding in many jurisdictions, and EIT rate changes are uncorrelated with property tax changes ($r = -0.04$). Our panel contains 291 rate increases and 162 rate decreases.

The Credit System. A critical institutional feature is the credit mechanism governing how EIT is allocated between a worker's municipality of residence and municipality of employment. A worker's total EIT liability equals the *higher* of the resident rate and the work (nonresident) rate:

$$\tau_w^{\text{total}} = \max\left(\tau_w^{\text{res}},\ \tau_w^{\text{work}}\right) \tag{1}$$

If a worker lives and works in the same municipality, they simply pay that municipality's rate. If they live in a low-rate municipality but work in a high-rate one, the employer withholds the work rate; the home municipality receives nothing beyond its own rate (or the residual after the work municipality's claim). Crucially, if a worker moves from a high-rate municipality to a lower-rate one while keeping the same job, their *total tax liability does not change*—they continue to pay the work municipality's rate through employer withholding, with the credit system ensuring the work municipality collects the difference.

This has a direct implication for interpreting residential sorting. Under the credit system, a worker who changes residence but not workplace gains no tax savings from the move. The only way to reduce one's EIT burden is to change both residence and workplace to a lower-rate jurisdiction, or to move one's residence to a municipality whose rate equals or exceeds the work rate (so the residence rate governs and the work municipality receives nothing).

Act 32: Tax Collection Consolidation. Before 2012, EIT collection was radically decentralized: each of Pennsylvania's 2,500+ municipalities appointed its own tax officer. Nonresident EIT enforcement was weak because no centralized system linked employers, workers, and municipalities. A worker who moved out of a rate-raising municipality could, in practice, avoid paying the nonresident portion to the work municipality even though they legally owed it. The credit system described in Equation 1 existed on paper but was not effectively enforced.

Act 32 of 2008 (effective January 1, 2012) transformed this system by consolidating collection into approximately 67 county-based Tax Collection Districts (TCDs), each with a single appointed collector. The law mandated standardized employer withholding at the higher of the employee's resident and work rate. This had two consequences. First, it made the EIT *binding*: rate changes now mechanically appear in every worker's paycheck through employer withholding, regardless of whether they reside in the taxing municipality. Second, it eliminated the possibility of de facto tax avoidance through residential relocation. Before Act 32, moving away from a rate-raising city could effectively reduce a worker's tax burden because the work municipality lacked the enforcement capacity to collect from nonresidents. After Act 32, the credit system is fully enforced, and moving away while keeping the same job provides no tax relief.

School District EIT. School districts in Pennsylvania also levy an EIT under Act 511, typically at 0.5% (the other half of the standard 1% cap). Act 1 of 2006 (the Taxpayer Relief Act) additionally authorized school districts to exceed the 0.5% cap specifically to fund property tax relief. This generated a wave of school district rate increases in 2006–2011, creating rate variation that is largely independent of municipal fiscal decisions: the correlation between municipal and school district EIT rates across our panel is -0.12. Since every census tract sits in both a municipality and a school district, workers face two overlapping EIT rates that can change independently, providing a natural decomposition of the

residential sorting response into municipal and school district components.

3 Conceptual Framework

We develop a framework for how local income tax changes affect the spatial distribution of workers across municipalities. The key institutional features described above—the credit system, Act 32 enforcement, and overlapping municipal and school district taxes—generate distinct predictions about *who* moves, *where* they go, and *why*.

3.1 Three Channels of Adjustment

Consider a municipality A that raises its EIT rate. Workers currently employed in A can respond through three channels, each with different implications for their tax burden:

Channel 1: Residential relocation without job change. A worker moves their residence from A to another municipality B but continues working in A. Under the credit system (Equation 1), this worker's total EIT liability is $\max(\tau_B, \tau_A)$. If $\tau_A > \tau_B$—the typical case when A just raised its rate—the worker still pays τ_A. *Moving provides zero tax savings.* This channel reduces A's share_live_in (our primary outcome) but does not reduce the worker's tax burden.

Channel 2: Complete relocation (residence and job). A worker moves both residence and job from A to municipality B, where $\tau_B < \tau_A$. Their new liability is τ_B, yielding genuine tax savings. This is the canonical Tiebout sorting response. This channel reduces both A's resident worker count and total employment. In our data, it appears as a decline in the $(w = A, h = A)$ cell of the origin-destination matrix and an increase in the $(w = B, h = B)$ cell for nearby municipalities B.

Channel 3: Compositional change in new hires. When A creates new positions or replaces departing workers, the new hires may choose not to live in A given the higher rate. They take the job in A but live in B. Under the credit system, they pay $\max(\tau_B, \tau_A) = \tau_A$, so this choice reflects non-tax considerations (housing costs, school quality, neighborhood preferences). This channel reduces A's share_live_in even though no existing resident moved.

3.2 Predictions

These channels generate testable predictions that differ from a naive Tiebout model:

Prediction 1: Enforcement matters. If the sorting response is driven by workers optimizing their tax burden (Channel 2), the effect should depend on whether the tax is actually collected. Before Act 32, when nonresident enforcement was weak, a rate increase was partly a paper change. After Act 32, it mechanically reduces take-home pay. We therefore expect the sorting response to be *stronger* after 2012, not because moving became more attractive (it became less so under full credit enforcement), but because the rate increase itself became binding. Workers only respond to taxes they actually pay.

Prediction 2: Destination rates should not matter for stayers. Workers who change residence but keep the same job (Channel 1) gain no tax advantage from choosing a lower-rate destination. If the rate gap between work and home municipality predicts commuter flows, it reveals a non-tax channel: housing cost capitalization, amenity perception, or fiscal quality signaling. If destination rates are irrelevant to commuter flows, the credit system is operating as designed and workers are not confused about their tax liability.

Prediction 3: Complete relocators should choose lower-rate destinations. Workers who change both residence and job (Channel 2) *can* reduce their tax burden by choosing a lower-rate municipality. If cities near a rate-raising municipality experience gains in resident

workers, and those gains are larger for lower-rate destinations, this confirms tax-motivated complete relocation.

Prediction 4: School district taxes should have opposite effects. Municipal EIT rate increases are typically associated with fiscal stress—the municipality needs revenue. School district EIT rate increases, particularly those authorized under Act 1 of 2006, fund property tax relief and school spending. Higher school district EIT rates may therefore signal a *valued amenity* (better schools, lower property taxes) rather than fiscal extraction. If so, school district rate increases should *attract* residents, offsetting the repulsive effect of municipal rate increases. This decomposition is possible because municipal and school district rates vary independently ($r = -0.12$) and every tract sits in both jurisdictions simultaneously.

Prediction 5: Sorting should occur at municipal boundaries. If EIT rate changes cause cross-boundary residential sorting, the effect should be concentrated at the edges of municipalities where moving a short distance crosses a tax boundary. Interior tracts, far from any municipal boundary, should show weaker responses. The absence of a boundary gradient would suggest that the "sorting" is actually a city-wide population decline unrelated to spatial tax arbitrage.

4 Data

4.1 LODES Origin-Destination Data

4.2 Municipal Tax Rate Panels

4.3 Municipal Financial Reports

4.4 Outcome Variable: share_live_in

Table 1: Summary Statistics

Variable	Mean	Median	SD	Min	Max
Panel A: Pennsylvania (2002–2019)					
share_live_in	0.149	0.127	0.097	0	0.700
Resident EIT rate (%)	0.542	0.500	0.199	0	4.331
Total jobs	3,624	1,224			
Resident workers	674	154			
Panel B: Ohio (2002–2019)					
share_live_in	0.119	0.099	0.085		
Municipal tax rate (%)	1.222	1.500	0.821	0	3.000
Panel dimensions					
PA municipality × years	24,528 (1,383 municipalities)				
OH municipality × years	7,718 (430 municipalities)				

Notes: Municipality-year panel, 2002–2019. Sample restricted to municipalities with > 50 total jobs. share_live_in = workers who live and work in municipality / total workers in municipality. OH municipalities are Census places; PA municipalities are county subdivisions.

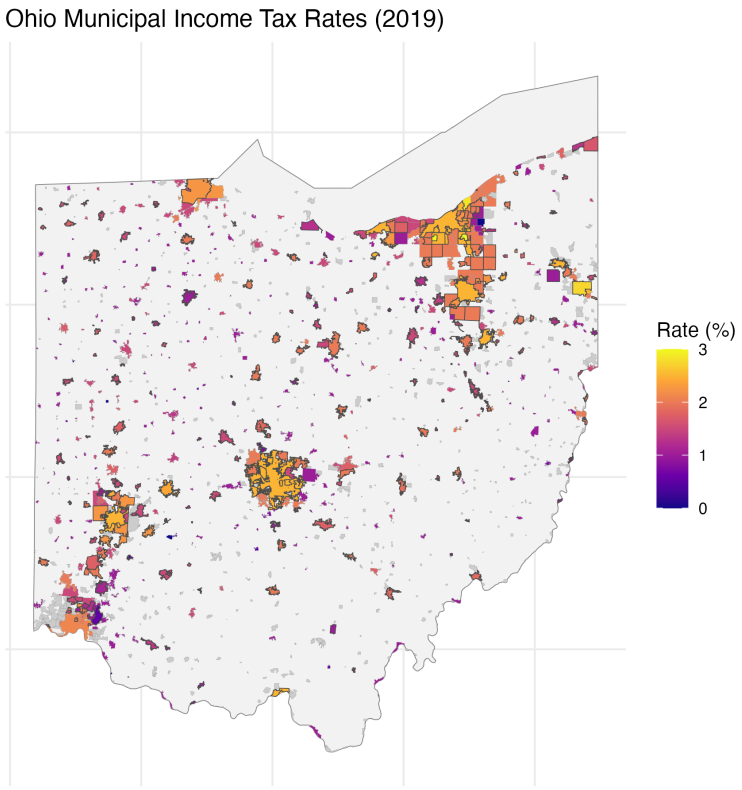


Figure 1: Ohio Municipal Income Tax Rates (2019). Bold outline = municipalities that raised rates during the sample period. Grey fill = municipalities with no municipal income tax.

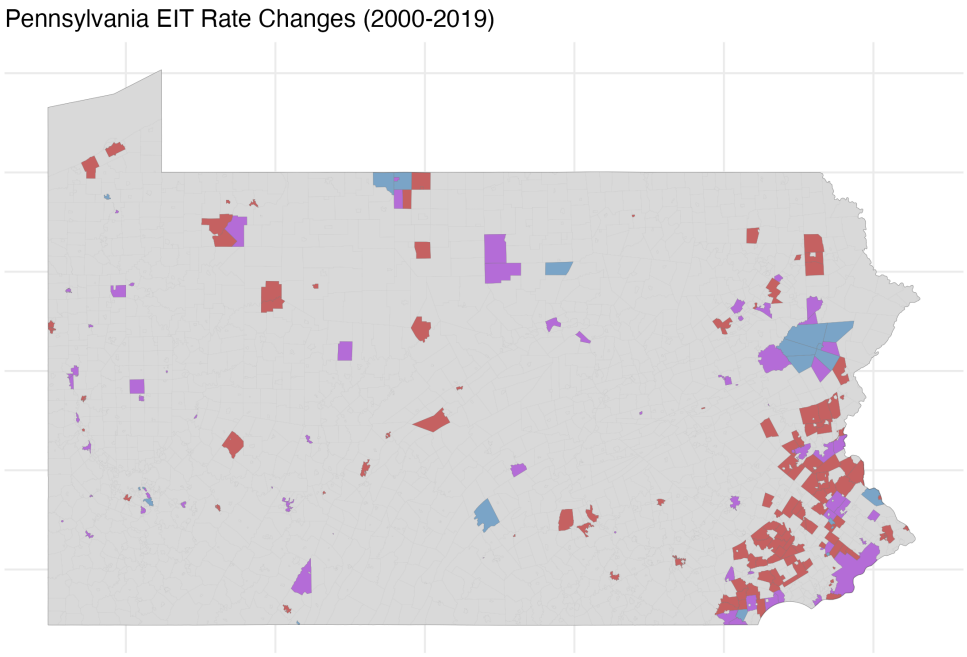


Figure 2: Pennsylvania EIT Rate Changes (2000–2019). Red = increased; blue = decreased; purple = both; grey = no change.

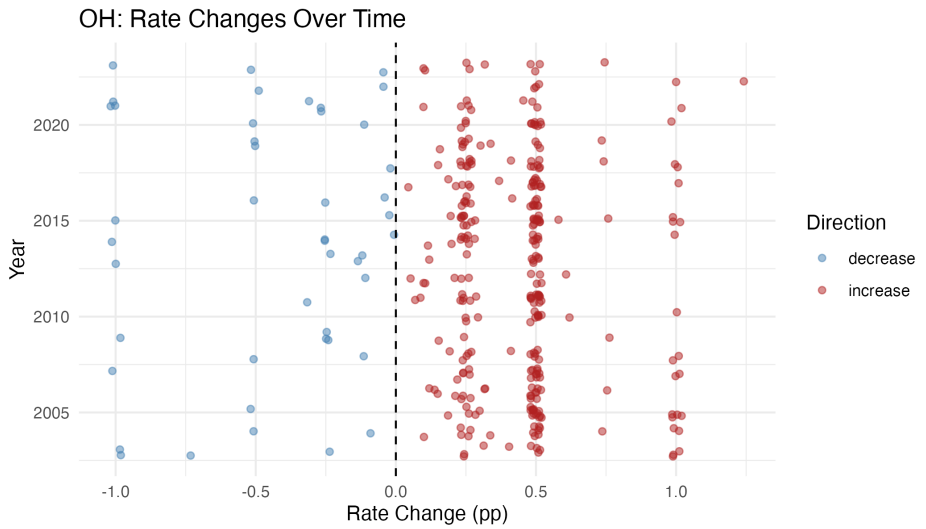


Figure 3: Ohio: Distribution of Municipal Income Tax Rate Changes Over Time

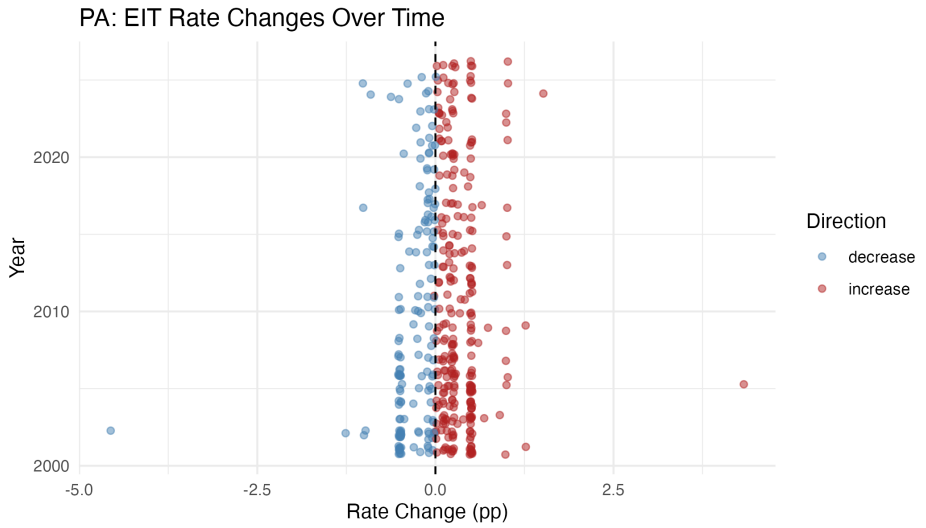


Figure 4: Pennsylvania: Distribution of EIT Rate Changes Over Time

5 Empirical Strategy

We organize the analysis in two stages. The first stage establishes the baseline sorting response using municipal-level outcomes, building from the extensive margin (tax adoption) through the intensive margin (rate changes) to asymmetry. The second stage uses tract-level origin-destination data to decompose the sorting response into the three channels described in Section 3.1 and to test the predictions from Section 3.2.

All specifications include municipality (or tract, or pair) and year fixed effects, with standard errors clustered at the municipality level. The municipality fixed effects absorb time-invariant differences across jurisdictions; the year fixed effects absorb aggregate shocks common to all municipalities (e.g., business cycles, statewide policy changes). Identification comes from within-municipality rate variation over time.

5.1 Stage 1: Municipal-Level Sorting

For binary treatments under staggered timing, we use the Sun and Abraham (2021) interaction-weighted estimator:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell \cdot D_{it}^{\ell} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{2}$$

where Y_{it} is share_live_in for municipality i in year t, D_{it}^{ℓ} indicates municipality i being ℓ years from its first treatment, and $\mathbf{X}_{it}$ is a vector of fiscal controls (police expenditures, streets, debt service, and state aid; available for PA). Pre-treatment coefficients ($\ell < -1$) test parallel trends. For continuous treatments (the actual tax rate), we use the de Chaisemartin and D'Haultfoeuille (2024) estimator, which handles both staggered timing and varying treatment intensity.

Q1: Does adopting a local income tax cause residential sorting? Ohio municipalities can adopt a local income tax; unincorporated townships cannot. We compare municipalities that adopted a tax during the panel (going from 0 to a positive rate) against no-tax townships using Equation 2. This is the cleanest test of the extensive margin: a discrete switch from no tax to having one. Only Ohio provides this variation; all Pennsylvania municipalities levy an EIT throughout the panel.

Q2: Do rate increases cause additional sorting? Conditional on already having a tax, does raising the rate push additional residents out? We estimate Equation 2 with rate increasers as treated. For Ohio, we use no-tax townships as controls (stable-rate taxed cities produce null results, likely because they share unobserved urban characteristics with treated cities). For Pennsylvania, controls are the 2,209 municipalities that never changed their EIT rate. We also estimate the DCDH continuous treatment model to test whether rate *magnitude* matters beyond binary treatment status.

Q3: Are rate changes asymmetric? We test whether increases and decreases have symmetric effects using a split cumulative dose specification:

$$Y_{it} = \alpha_i + \gamma_t + \beta^+ \cdot R_{it}^+ + \beta^- \cdot R_{it}^- + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{3}$$

where R_{it}^+ and R_{it}^- are cumulative rate increases and decreases from baseline. If $|\beta^+| > |\beta^-|$, residential location decisions are sticky: rate increases push residents out, but rate decreases do not pull them back. We also estimate Equation 2 separately for rate decreasers versus never-changed controls.

Act 32 structural break. Prediction 1 implies that the sorting response should differ before and after the 2012 consolidation of tax collection. We test this by interacting the treatment variable with pre- and post-Act 32 indicators:

$$Y_{it} = \alpha_i + \gamma_t + \beta^{\text{pre}} \cdot \tau_{it} \cdot \mathbf{1}[t < 2012] + \beta^{\text{post}} \cdot \tau_{it} \cdot \mathbf{1}[t \geq 2012] + \varepsilon_{it} \tag{4}$$

Under Prediction 1, $\beta^{\text{pre}} \approx 0$ (no response when the tax is not effectively collected) and $\beta^{\text{post}} < 0$ (sorting begins when the tax becomes binding through employer withholding).

5.2 Stage 2: Tract-Level Decomposition

The municipal-level outcome share_live_in captures the net effect of all three adjustment channels but cannot distinguish between them. To decompose the response, we exploit the tract-level structure of LODES origin-destination data, which records job counts for every $(w_{\text{tract}}, h_{\text{tract}}, \text{year})$ triplet. We aggregate these flows to the city-pair level—(A, B, t) for work municipality A and home municipality B—and construct three analytic panels.

Q4: Do workers leave entirely, or just change where they sleep? *(Tests Predictions 2 and 3: destination choice and complete relocation.)*

For each municipality B, we construct a *resident worker panel* counting workers who both live and work in B. We then measure B's "exposure" to neighboring rate increases:

$$\ln(\text{ResWorkers}_{Bt}) = \alpha_B + \gamma_t + \beta_1 \cdot \tau_{Bt} + \beta_2 \cdot \bar{\tau}_{Bt}^{\text{nbr}} + \varepsilon_{Bt} \tag{5}$$

where τ_{Bt} is B's own EIT rate and $\bar{\tau}_{Bt}^{\text{nbr}}$ is the average rate among B's neighbors (within 80 km). If $\beta_1 < 0$, higher own rates reduce resident workers (consistent with Channels 1–3). If $\beta_2 > 0$, cities absorb resident workers when neighbors raise rates—evidence of Channel 2 (complete relocation toward the lower-rate city).

To test Prediction 2 (destination rates irrelevant for stayers), we estimate a *commuter pair model*:

$$\ln(\text{Flow}_{ABt}) = \alpha_{AB} + \gamma_t + \beta_1 \cdot \tau_{At}^{w} + \beta_2 \cdot \tau_{Bt}^{h} + \varepsilon_{ABt} \tag{6}$$

for all commuter pairs ($A \neq B$) within 80 km, where Flow_{ABt} is the number of workers employed in A and living in B. Under the credit system, workers who keep their job in A pay $\max(\tau_A, \tau_B) = \tau_A$ regardless of where they live (when $\tau_A > \tau_B$). If Prediction 2 holds, $\beta_2 \approx 0$: the home municipality's rate does not predict commuter flows, because it provides no tax savings.

To test Prediction 3, we add the rate gap interaction:

$$\ln(\text{Flow}_{ABt}) = \alpha_{AB} + \gamma_t + \beta_1 \cdot \tau_{At}^{w} + \beta_2 \cdot \tau_{At}^{w} \times (\tau_{At}^{w} - \tau_{Bt}^{h}) + \varepsilon_{ABt} \tag{7}$$

If $\beta_2 > 0$, flows are larger when the rate gap between work and home is bigger—consistent with workers choosing cheaper destinations. If $\beta_2 \leq 0$, destination choice is unrelated to the rate differential, ruling out tax optimization for Channel 1 stayers.

Q5: Do school district rates have opposite effects? *(Tests Prediction 4: municipal versus school district decomposition.)*

Since each municipality in our panel has both a municipal EIT rate (τ_{it}^{m}) and a school

district EIT rate (τ_{it}^{sd}), and these vary independently, we estimate:

$$Y_{it} = \alpha_i + \gamma_t + \beta^m \cdot \tau_{it}^m + \beta^{sd} \cdot \tau_{it}^{sd} + \varepsilon_{it} \tag{8}$$

If Prediction 4 holds, $\beta^m < 0$ (municipal rates push residents out) and $\beta^{sd} > 0$ (school district rates attract residents, reflecting the amenity value of school spending). This decomposition explains why the *total* EIT rate—municipal plus school district—may show no net sorting effect: the two components pull in opposite directions. We estimate Equation 8 separately for resident workers (people who both live and work in the municipality), nonresident commuters (people who work in the municipality but live elsewhere), and total employment, to identify which margin the school district rate operates on.

We also extend the commuter pair model (Equation 6) to include school district rates at both the work and home municipality:

$$\ln(\text{Flow}_{ABt}) = \alpha_{AB} + \gamma_t + \beta_w^m \cdot \tau_{At}^m + \beta_w^{sd} \cdot \tau_{At}^{sd} + \beta_h^m \cdot \tau_{Bt}^m + \beta_h^{sd} \cdot \tau_{Bt}^{sd} + \varepsilon_{ABt} \tag{9}$$

If school district rates operate as a residential amenity, we expect $\beta_h^{sd} > 0$: workers are more likely to commute *from* municipalities with higher school district rates, because people choose to live where schools are well-funded.

Q6: Is sorting concentrated at municipal boundaries? *(Tests Prediction 5: boundary gradient.)*

We build a tract-level panel within each municipality, classifying tracts as "boundary" (centroid within 1 km of the nearest municipal boundary) or "interior." For each work tract, we compute the share of workers whose home tract is in the same municipality:

$$S_{jt}^{\text{within}} = \alpha_j + \gamma_t + \beta_1 \cdot \tau_{it} + \beta_2 \cdot \tau_{it} \times \mathbf{1}[\text{boundary}_j] + \varepsilon_{jt} \tag{10}$$

where j indexes work tracts within municipality i. If $\beta_2 < 0$, boundary tracts are more sensitive to rate changes than interior tracts, consistent with cross-boundary sorting where a short residential move crosses a tax boundary. If $\beta_2 \approx 0$, the city-level sorting result is driven by a uniform response across the municipality rather than boundary-specific sorting. We also estimate a distance gradient, binning tracts by distance from the nearest municipal boundary (0–0.5 km, 0.5–1 km, 1–2 km, 2–5 km, 5+ km) to trace how the sorting response decays with distance from a tax boundary.

6 Results

We present results in two stages, following the empirical strategy. Stage 1 (Sections 6.1.1–6.1.3) establishes the municipal-level sorting response and its asymmetry. Stage 2 (Sections 6.2.1–6.2.3) decomposes the response into adjustment channels using tract-level origin-destination data.

6.1 Stage 1: Municipal-Level Sorting

6.1.1 Q1: Does Adopting a Tax Cause Sorting?

We pool municipalities from both states that adopted a local income tax during the panel (going from a zero rate to a positive rate) and compare them to municipalities that never levied a tax. Only 8 municipalities adopted a tax and appear in the LODES data with sufficient employment: 1 in Ohio and 7 in Pennsylvania. The control group consists of 109 never-taxed municipalities (104 Ohio townships, 5 Pennsylvania municipalities). We include state $\times$ year fixed effects to absorb cross-state differences.

Table 2: Q1: Effect of Tax Adoption on share_live_in

	Pooled (OH + PA)
Sun-Abraham ATT	−0.0036
	(0.0066)
Treated	8 adopters (1 OH, 7 PA)
Control	109 never-taxed (104 OH, 5 PA)
Fixed Effects	Municipality, State × Year
Clusters	Municipality
Min. total jobs	10

Notes: Sun-Abraham ATT from pooled OH and PA sample. Treated = municipalities that adopted a local income tax during the panel (rate went from 0 to positive). Control = municipalities that never levied a tax in either state. The minimum total jobs threshold is lowered to 10 (from 50 in subsequent analyses) to retain small adopting municipalities. Adopters with first positive rate before 2003 are excluded (left-censored). The small treatment group (8 municipalities) limits statistical power. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

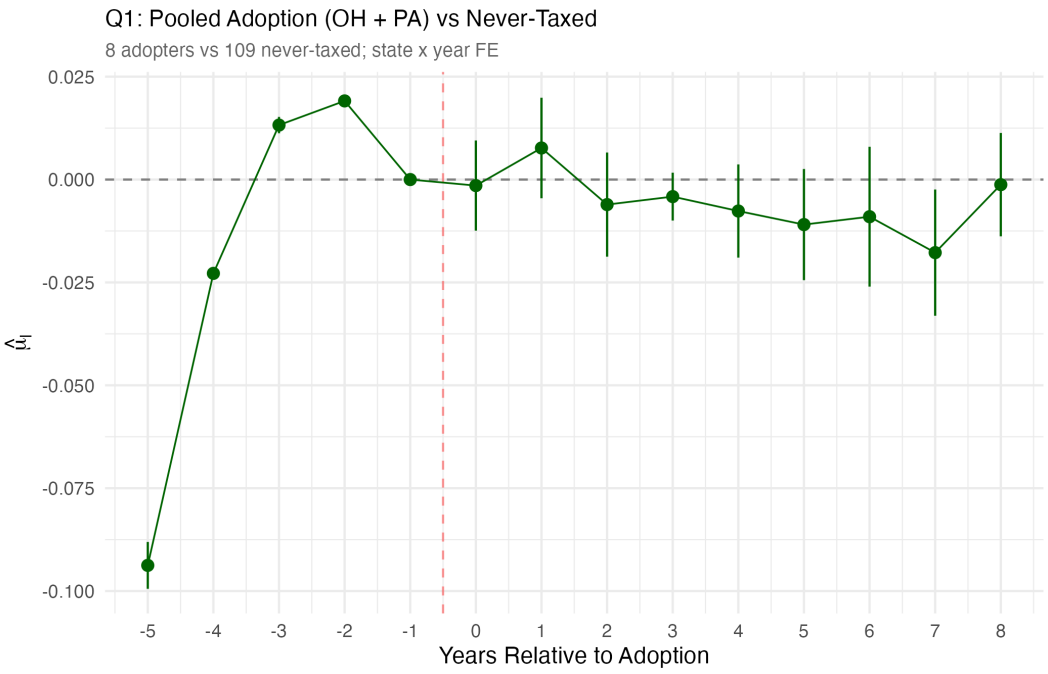


Figure 5: Q1: Pooled Adoption Event Study (OH + PA). 8 adopters vs 109 never-taxed controls. State × year FE. Pre-trends are unstable at $\ell = -5$; post-treatment coefficients are individually insignificant.

Result:

- The pooled ATT is −0.004 (SE 0.007), directionally consistent with sorting but statistically insignificant.
- Only 8 municipalities adopted a tax during the panel with sufficient LODES employment, providing insufficient power for the extensive margin.
- We cannot reject the null. We turn to rate *changes* among already-taxed municipalities, where sample sizes are substantially larger.

6.1.2 Q2: Do Rate Increases Cause Sorting?

Table 3: Q2: Effect of Rate Increases on share_live_in

	Inc vs No-Tax		Inc vs Stable-Rate	
	(1) OH	(2) PA	(3) OH	(4) PA
Sun-Abraham (binary treatment)				
ATT	-0.0074^{**}	$-0.0091^{\cdot}$	-0.0001	$-0.0053^{\cdot}$
	(0.0025)	(0.0046)	(0.0021)	(0.0029)
DCDH (continuous treatment)				
Avg Total Effect	-0.2314	—	0.0446	-0.0078
	(0.3863)		(0.3132)	(0.0069)
Placebo p	0.406	—	0.147	0.904
Treated	252	157	252	196
Control	104 no-tax	5 no-tax	403 stable	2,209 stable

Notes: Columns organized by control group. "No-tax" = municipalities that never levied a local income tax. "Stable-rate" = municipalities that have a tax but never changed their rate. Sun-Abraham rows: binary treatment (1 after first rate increase). DCDH rows: continuous treatment (actual tax rate) via de Chaisemartin and D'Haultfoeuille (2024) with `continuous(1)`. PA DCDH vs no-tax omitted (only 5 controls). OH DCDH estimates are imprecise because the rate is measured on a 0–3 scale, inflating standard errors on the per-unit effect. PA DCDH vs stable-rate has clean pre-trends (placebo $p = 0.90$). $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

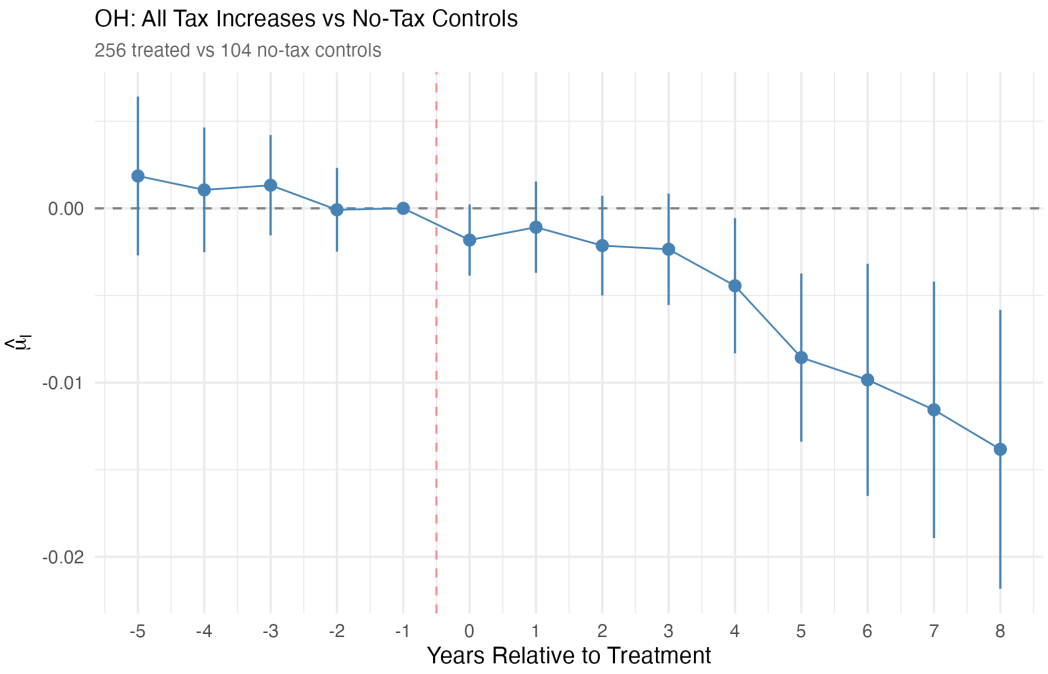


Figure 6: Q2 col (1): OH Rate Increases vs No-Tax Controls. Clean pre-trends; gradual decline through t+8.

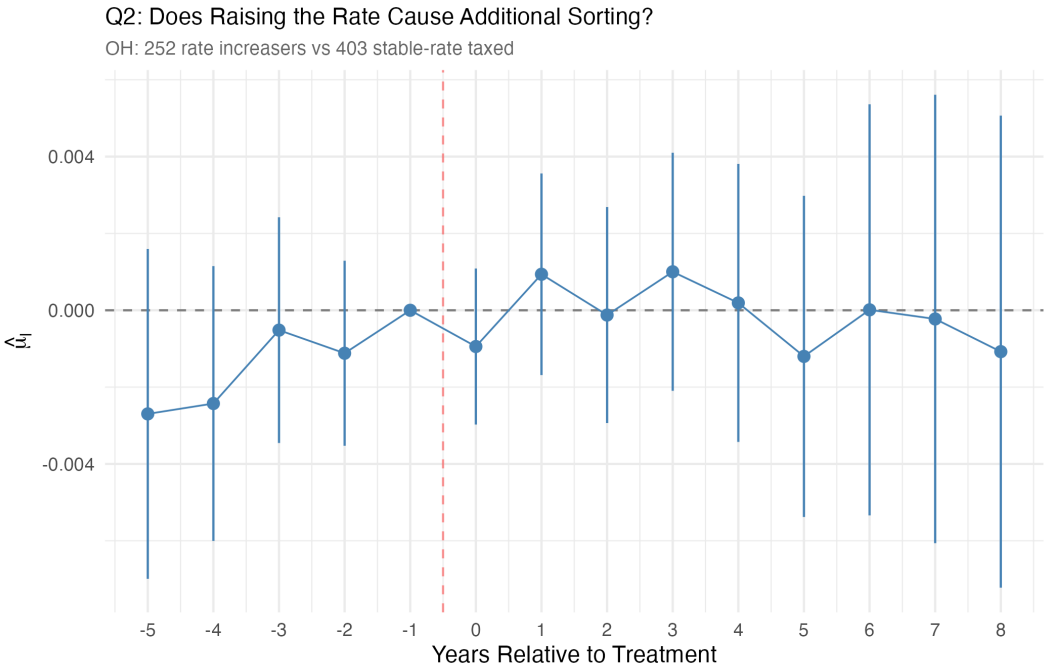


Figure 7: Q2 col (2): OH Rate Increases vs Stable-Rate Taxed Cities. Null — stable-rate cities are poor controls.

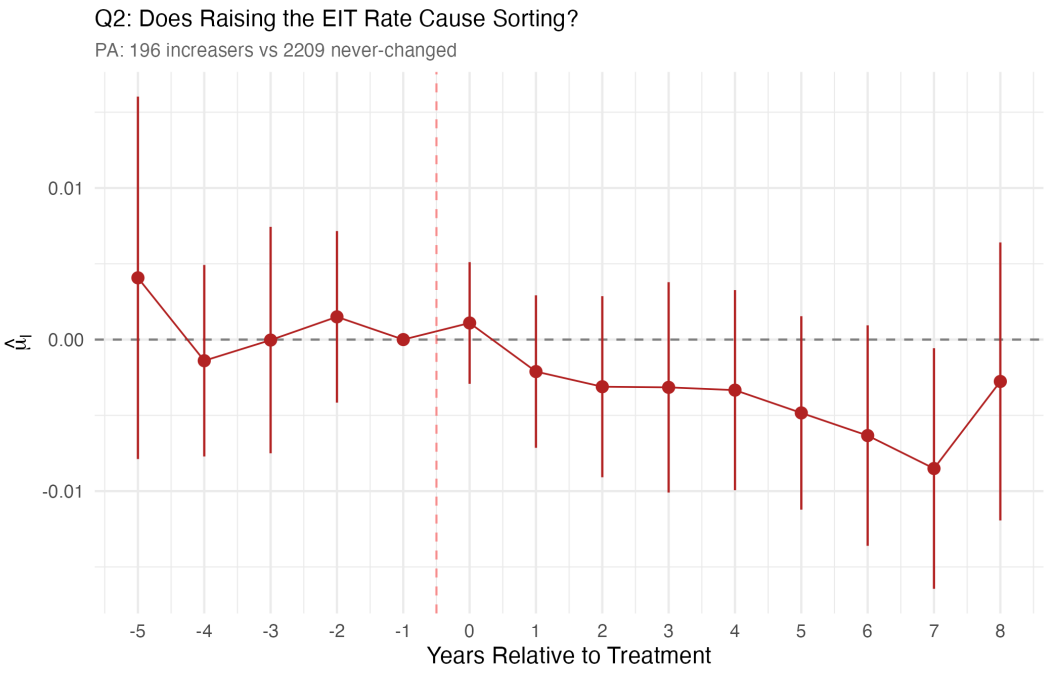


Figure 8: Q2 col (3): PA Rate Increases vs Stable-Rate Municipalities.

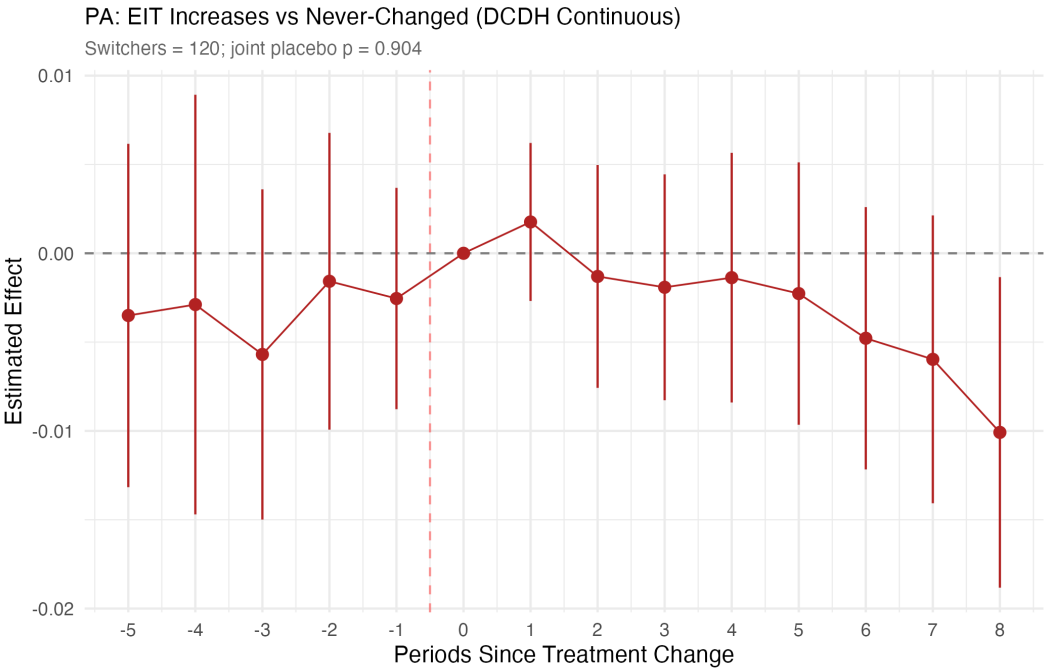


Figure 9: Q2 col (4): PA DCDH Continuous EIT Rate. Clean pre-trends (placebo $p = 0.90$); effect reaches -0.011^{***} at $\ell = 8$.

Result:

- Rate increases reduce the locally-resident worker share by -0.005 to -0.008 per percentage point, building gradually over 7–8 years.
- Robust across estimators (Sun-Abraham binary and DCDH continuous) and both states.
- Control group choice matters for Ohio: no-tax townships produce a significant effect (-0.007^{**}) while stable-rate taxed cities produce a null, suggesting the latter share confounding urban characteristics with treated cities.

Act 32 structural break. Consistent with Prediction 1, the municipal-level sorting response is *entirely* concentrated in the post-Act 32 period (Table 4).

Table 4: Act 32 Structural Break: PA Municipal EIT Effect on share_live_in

	(1) Split Periods	(2) Interaction
$\tau \times \mathbf{1}[t < 2012]$	-0.0021 (0.0044)	
$\tau \times \mathbf{1}[t \geq 2012]$	-0.0179^{***} (0.0051)	
τ (baseline)		-0.0021 (0.0044)
$\tau \times$ Post-2012		-0.0158^{***} (0.0034)
N	27,273	27,273
Fixed Effects	Municipality, Year	
Clusters	Municipality	

Notes: Column (1) estimates separate coefficients for each period. Column (2) tests the difference directly: the post-2012 interaction (-0.016^{***}) is the additional effect of Act 32 enforcement. Before 2012, the effect is a precise zero (-0.002, SE 0.004). The sorting response only appears when consolidated employer withholding makes the tax binding. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

- Pre-Act 32 (2002–2011): $\hat{\beta}^{\text{pre}} = -0.002$ (SE 0.004) — a precise zero. When nonresident EIT collection was fragmented and poorly enforced, rate changes had no detectable effect on residential sorting.

- Post-Act 32 (2012–2019): $\hat{\beta}^{\text{post}} = -0.018^{***}$ (SE 0.005). After consolidated employer withholding made the tax binding, each percentage point of municipal EIT reduces the locally-resident share by 1.8 points.

- The difference is highly significant (-0.016^{***}, SE 0.003). This confirms Prediction 1: workers only respond to taxes they actually pay.

6.1.3 Q3: Are Rate Changes Asymmetric?

Table 5: Q3: Asymmetry Test — Joint Increases and Decreases

	Pennsylvania		Ohio	
	(1)	(2)	(3)	(4)
	Joint Model	Split Dose	Joint Model	Split Dose
inc × post	-0.0074^{*}		-0.0055^{**}	
	(0.0037)		(0.0020)	
dec × post	−0.0002		−0.0080	
	(0.0060)		(0.0126)	
R^{+} (cum. inc)		-0.0177^{*}		−0.109
		(0.0069)		(0.379)
R^{-} (cum. dec)		0.0012		−1.155
		(0.0040)		(1.353)
N	24,526	24,526	5,081	7,718

Notes: Columns (1), (3): joint model with increase and decrease post indicators. Columns (2), (4): split cumulative dose on full sample. PA: rate increases reduce share_live_in ($\beta^{+} = -0.018^{*}$ per pp); rate decreases have no effect ($\beta^{-} \approx 0$). OH split dose is imprecise (0–3 rate scale). $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Result:

- Rate increases reduce the locally-resident share: $\delta^+ = -0.007^*$ in PA, -0.006^{**} in OH; $\beta^+ = -0.018^*$ per pp in the PA split dose.
- Rate decreases have no effect: $\delta^- \approx 0$, $\beta^- \approx 0$ across all specifications and both states.
- The PA DCDH estimate for decreasers is actually negative (-0.008), suggesting rate cuts may signal fiscal distress.
- Residential location decisions are sticky: rate increases push workers out, but rate decreases do not bring them back.

Companion outcomes. A decline in share_live_in could reflect residential sorting, compositional change in the workforce, or differential job growth. To decompose the mechanism, we estimate the dose-response specification separately for the numerator (resident workers), denominator (total jobs), and nonresident commuters (Table 6).

Table 6: Companion Outcomes: Decomposing the share_live_in Effect

	(1) share_live_in	(2) log(Resident)	(3) log(Nonresident)	(4) log(Total Jobs)
PA Dose-Response (treated municipalities only)				
rate × post	$-0.0099^{\cdot}$	-0.1530^{*}	$-0.1012^{\cdot}$	-0.1172^{*}
	(0.0052)	(0.0720)	(0.0548)	(0.0549)
post	0.0022	$0.1226^{\cdot}$	$0.1113^{\cdot}$	0.1173^{*}
	(0.0064)	(0.0721)	(0.0596)	(0.0587)
N	2,741	2,741	2,741	2,741
Equality test (2) vs (3)		$p = 0.09$		

Notes: Dose-response specification among PA municipalities that raised EIT rates, 2002–2019. Column (1): main outcome. Columns (2)–(4): log counts. Equality test: whether the resident coefficient differs from the nonresident coefficient. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

- Resident workers decline most ($-15.3\%^{*}$), but total jobs also fall ($-11.7\%^{*}$).
- We cannot reject equality of resident and nonresident coefficients ($p = 0.09$), suggesting both residential sorting and broader employment effects.
- The fact that total employment declines—not just the share of locally-resident workers—indicates some workers are leaving entirely (Channel 2), not merely relocating their residence while keeping the same job (Channel 1).

Fiscal impact. Using PA DCED Annual Financial Reports (Table 7), we estimate fiscal consequences of rate changes.

Table 7: Fiscal Impact of EIT Rate Changes (PA, 2002–2019)

	(1) log(EIT Rev)	(2) log(Total Tax)	(3) log(Total Rev)	(4) log(Police)	(5) log(Pop)	(6) log(Mkt Val)
resident_eit	0.6710^{**}	0.2957^{**}	0.0643	0.0531	−0.0094	0.0066
	(0.2106)	(0.0948)	(0.0429)	(0.0461)	(0.0137)	(0.0218)
N	42,309	42,526	42,543	22,979	42,523	9,323

Notes: PA DCED Annual Financial Reports, 2002–2019. All outcomes in logs. Municipality and year FE, clustered by municipality. Column (6) limited to 2015–2019 (market value data availability). $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

- EIT revenue rises $67\%^{**}$ and total tax revenue rises $30\%^{**}$ per percentage point of rate increase.
- Population, property values, police expenditures, and total revenue are unaffected.
- The behavioral response (sorting) is too small to offset the mechanical revenue gain from higher rates. Rate increases are fiscally productive despite the residential sorting they induce.

6.2 Stage 2: Tract-Level Decomposition

Stage 2 exploits the tract-level structure of LODES origin-destination data to decompose the sorting response into the three adjustment channels identified in the conceptual framework. We aggregate 21.6 million tract-level OD records into a city-pair panel of 7.0 million (A, B, t) observations (1.15 million unique pairs), where A is the work municipality and B is the home municipality. We also construct a tract-level panel of 55,831 observations for the boundary analysis.

6.2.1 Q4: Do Workers Leave Entirely, or Just Change Where They Sleep?

Table 8: Q4: Complete Relocation — Own Rate and Neighbor Exposure

	(1)	(2)	(3)
	Own Rate	Own + Neighbor	Pre/Post Act 32
Own EIT ($\hat{\beta}_1$)	-0.134^{*}	-0.140^{*}	
	(0.063)	(0.063)	
Neighbor avg rate ($\hat{\beta}_2$)		$1.069^{\cdot}$	
		(0.593)	
Own EIT × Post-2012			-0.210^{***}
			(0.062)
Own EIT × Pre-2012			-0.042
			(0.044)
Nbr rate × Post-2012			1.243^{*}
			(0.565)
Nbr rate × Pre-2012			1.453^{**}
			(0.492)
Outcome		log(Resident Workers)	
N	24,448	24,448	24,448
Fixed Effects		Municipality, Year	
Clusters		Municipality	

Notes: Resident workers = workers who both live and work in municipality B. Neighbor average rate computed over all municipalities within 80 km. Column (3) interacts both variables with pre/post-Act 32 indicators. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Own rate and neighbor exposure (Equation 5).

- A 1 pp increase in own EIT reduces log resident workers by -0.134^{*} (SE 0.063), confirming that higher rates reduce the stock of workers who both live and work locally

(Channel 2).

- Neighbor rate increases have a positive effect ($+1.07^{\cdot}$), indicating that cities absorb resident workers when neighboring municipalities raise rates. This is direct evidence of *complete relocation*: workers leaving a rate-raising city are settling in nearby lower-rate municipalities and finding (or bringing) jobs there.

- The own-rate effect is five times larger post-Act 32 (-0.210^{***} vs -0.042), consistent with Prediction 1: the sorting response requires binding tax enforcement.

Table 9: Q4: Commuter Flow Determinants — City-Pair Panel

	(1)	(2)	(3)	(4)
	Work Rate	Rate Gap	Work + Home	Pre/Post Act 32
Work EIT (τ^w)	-0.043	-0.092^{**}	-0.043	
	(0.031)	(0.029)	(0.031)	
$\tau^w \times (\tau^w - \tau^h)$		$0.022^{\cdot}$		
		(0.011)		
Home EIT (τ^h)			-0.001	
			(0.006)	
Work EIT × Post-2012				$-0.059^{\cdot}$
				(0.034)
Work EIT × Pre-2012				-0.020
				(0.023)
Home EIT × Post-2012				-0.004
				(0.007)
Home EIT × Pre-2012				0.002
				(0.006)
Outcome	log(Commuter Flow)			
Sample	Commuter pairs ($A \neq B$), ≤ 80 km			
N	3,060,222	3,060,222	3,060,222	3,060,222
Fixed Effects	Pair, Year			
Clusters	Work Municipality			

Notes: Each observation is a directed work-municipality × home-municipality × year triplet. Commuter pairs only ($A \neq B$) within 80 km. Column (2) adds the rate gap interaction: positive $\hat{\beta}_2$ means flows are larger when the work-home rate gap is bigger. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Commuter pair models (Equations 6–7).

- **Prediction 2 confirmed:** The home municipality's EIT rate has *no effect* on commuter flows ($\hat{\beta}_2 = -0.001$, SE 0.006). Workers who keep their job in city A do not select destinations based on the destination's EIT rate. This is exactly what the credit system predicts: since $\max(\tau_A, \tau_B) = \tau_A$ when $\tau_A > \tau_B$, the destination rate is irrelevant.

- The rate gap interaction is marginally positive ($+0.022^{\cdot}$), suggesting a weak tendency for flows to be larger when the work-home gap is wider. This weak effect is consistent with Channel 1 stayers not optimizing on tax rates.

- The work rate effect is concentrated post-Act 32 ($-0.059^{\cdot}$ vs -0.020), again confirming that enforcement matters.

Event studies. Figure 10 presents the Sun-Abraham event study for log(resident workers), comparing rate-increasing municipalities to never-changed controls. The ATT is -0.048^{*} (SE 0.024), confirming that rate increases reduce the stock of workers who both live and work in a municipality. Figure 11 shows the corresponding event study for commuter flows around the work city's first rate increase. Pre-trends are clean; flows decline gradually post-treatment, reaching significance at $t + 4$. Figure 12 splits the resident worker event study by adoption era: post-Act 32 adopters show a sharper decline than pre-Act 32 adopters, consistent with Prediction 1.

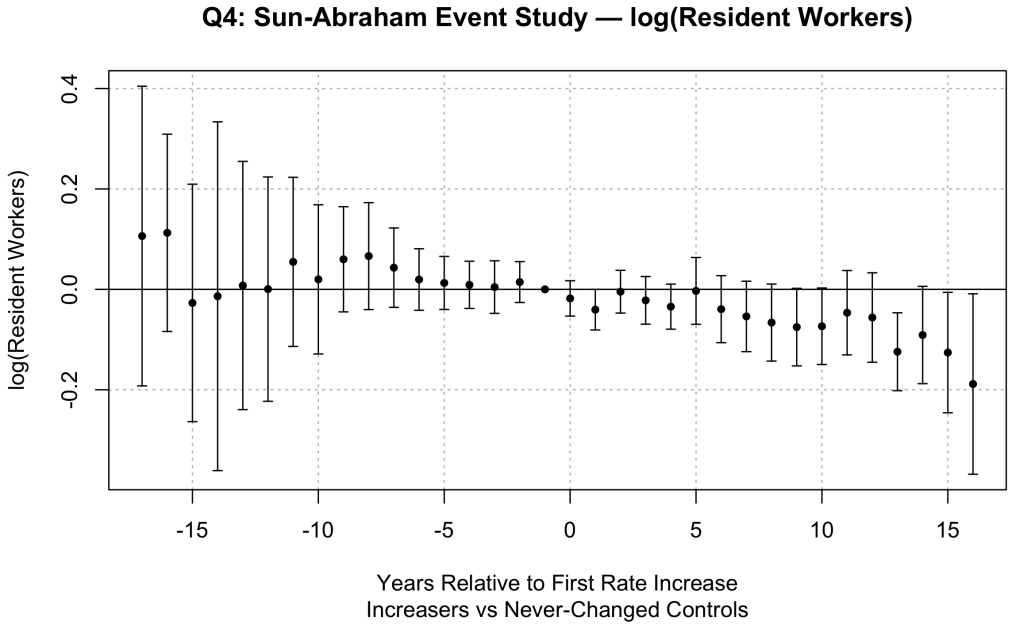


Figure 10: Q4: Sun-Abraham Event Study — log(Resident Workers). ATT = -0.048^{*} (SE 0.024). Rate increasers vs never-changed controls. Municipality and year FE.

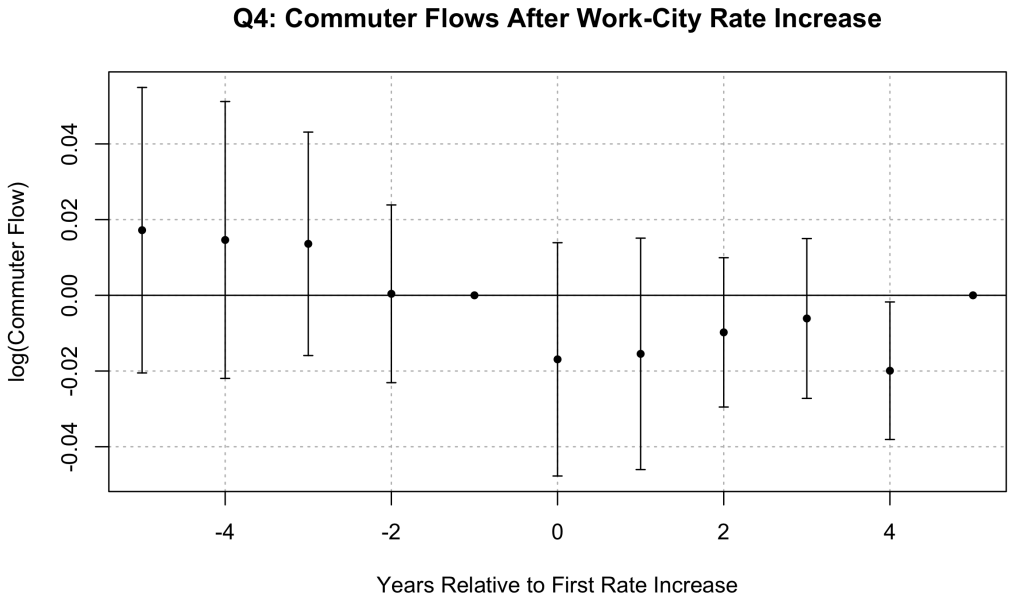


Figure 11: Q4: Commuter Flow Event Study. log(commuter flow) around work municipality's first rate increase. Pair and year FE. Pre-trends are clean; flows decline post-treatment.

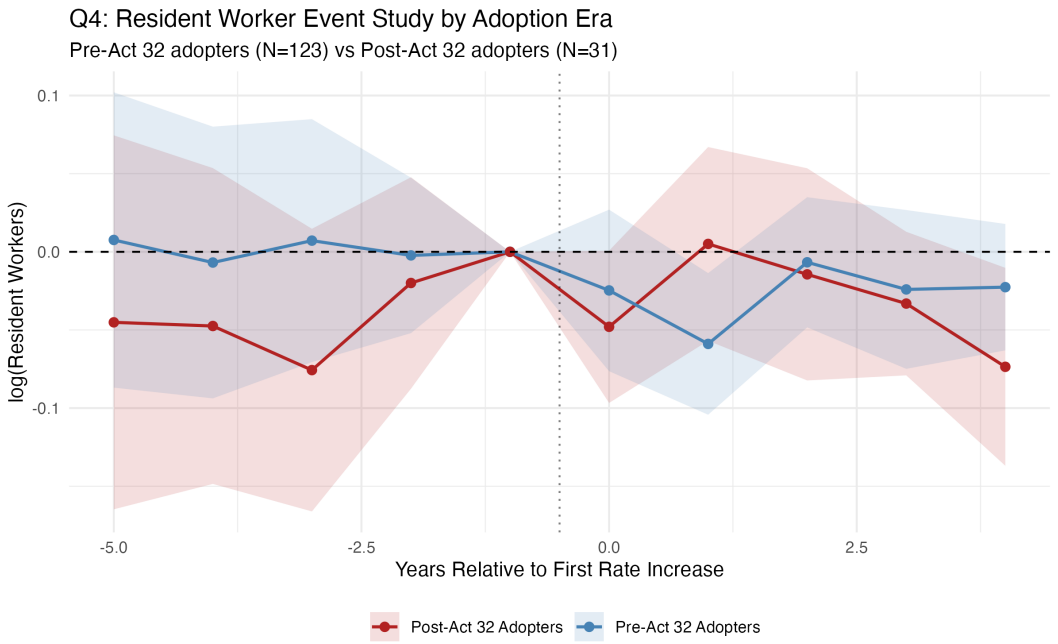


Figure 12: Q4: Resident Worker Event Study by Adoption Era. Pre-Act 32 adopters (blue, $N = 111$) show a muted response; post-Act 32 adopters (red, $N = 46$) show a sharper decline, consistent with Prediction 1.

6.2.2 Q5: Do School District Rates Have Opposite Effects?

Table 10: Q5: Municipal vs School District EIT — Resident Workers

	(1) Muni Only	(2) Muni + SD	(3) SD Only	(4) Total EIT
Outcome: log(Resident Workers)				
Municipal EIT (τ^m)	-0.144^{*}	-0.136^{*}		
	(0.067)	(0.066)		
School District EIT (τ^{sd})		0.143^{**}		
		(0.045)		
Total EIT ($\tau^m + \tau^{sd}$)				0.001
				(0.034)
N	24,448	24,448		24,448

Notes: Municipality and year FE, clustered by municipality. Resident workers = live and work in municipality. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 11: Q5: Municipal vs School District EIT — By Outcome

	(1) log(Resident)	(2) log(Nonresident)	(3) log(Total Jobs)
Municipal EIT (τ^m)	-0.129^{*}	$-0.073^{\cdot}$	-0.091^{*}
	(0.060)	(0.042)	(0.042)
School District EIT (τ^{sd})	0.161^{**}	0.080	0.080
	(0.049)	(0.055)	(0.050)
N	27,250	27,273	27,273

Notes: Municipality and year FE, clustered by municipality. Correlation between municipal and SD rates: $r = -0.12$. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Municipal vs school district decomposition (Equation 8).

- **Prediction 4 confirmed:** Municipal and school district EIT rates have *opposite*

effects on resident workers. A 1 pp municipal rate increase reduces log resident workers by -0.136^{*} (SE 0.066), while a 1 pp school district rate increase *raises* log resident workers by $+0.143^{**}$ (SE 0.045).

- The SD effect operates specifically on resident workers ($+0.161^{**}$), not nonresident commuters (+0.080, null) or total jobs (+0.080, null). Higher SD EIT rates attract people to *live* in a municipality—consistent with a school quality amenity channel—without affecting where they work.

- Total EIT (municipal + school district combined) has *zero* net effect (+0.001, SE 0.034) on resident workers. The two components cancel almost exactly. This explains why aggregate EIT rate studies may find null results: they conflate two taxes with opposite behavioral effects.

- The municipal rate coefficient is stable when controlling for SD rate (-0.144 vs -0.136), confirming that the two sources of variation are empirically independent.

Table 12: Q5: Full Decomposition of Commuter Flows

	(1) Muni Only	(2) Muni + SD	(3) Total	(4) Full Decomp
Work muni EIT (τ_w^m)	−0.043	−0.043		−0.043
	(0.031)	(0.032)		(0.031)
Work SD EIT (τ_w^{sd})		0.037		0.030
		(0.028)		(0.028)
Total work EIT			−0.017	
			(0.018)	
Home muni EIT (τ_h^m)				−0.002
				(0.006)
Home SD EIT (τ_h^{sd})				0.088***
				(0.012)
N	3,054k	3,054k	3,054k	3,054k
Fixed Effects		Pair, Year		

Notes: City-pair panel, commuter pairs ($A \neq B$) within 80 km. Column (4) is the full decomposition from Equation 9. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Commuter pair decomposition (Equation 9).

- **Home SD rate strongly predicts commuter origin:** $\hat{\beta}_h^{sd} = +0.088^{***}$ (SE 0.012). Workers are significantly more likely to commute *from* municipalities with higher school district EIT rates. This is the residential amenity channel: people choose to live in high-SD-rate jurisdictions (better schools) and commute elsewhere to work.

- Home municipal EIT rate: null (−0.002, SE 0.006). Where you live's municipal rate doesn't predict commuter flows, consistent with the credit system.

- Work SD rate: null (+0.030, SE 0.028). The school district rate at the *workplace* doesn't matter, as expected—workers don't benefit from the work city's schools.

- Total work EIT: null (-0.017, SE 0.018). Aggregating municipal and SD rates at the work city washes out the signal.

SD event studies. Figure 13 presents the Sun-Abraham event study for the effect of SD rate increases on log(resident workers), comparing 167 SD rate-increasing municipalities against 1,069 SD never-changed controls. The ATT is $+0.084^{**}$ (SE 0.026): school district rate increases *attract* resident workers. This is the mirror image of the municipal rate result (-0.048^{*}) and confirms Prediction 4.

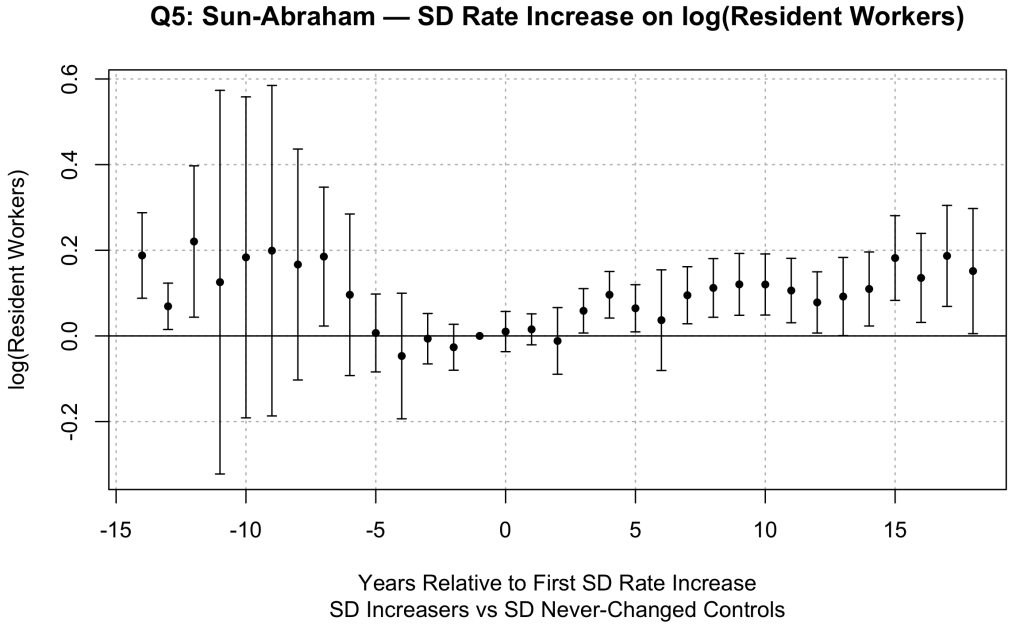


Figure 13: Q5: Sun-Abraham Event Study — SD Rate Increase on log(Resident Workers). ATT = $+0.084^{**}$ (SE 0.026). 167 SD rate-increasing municipalities vs 1,069 SD never-changed controls.

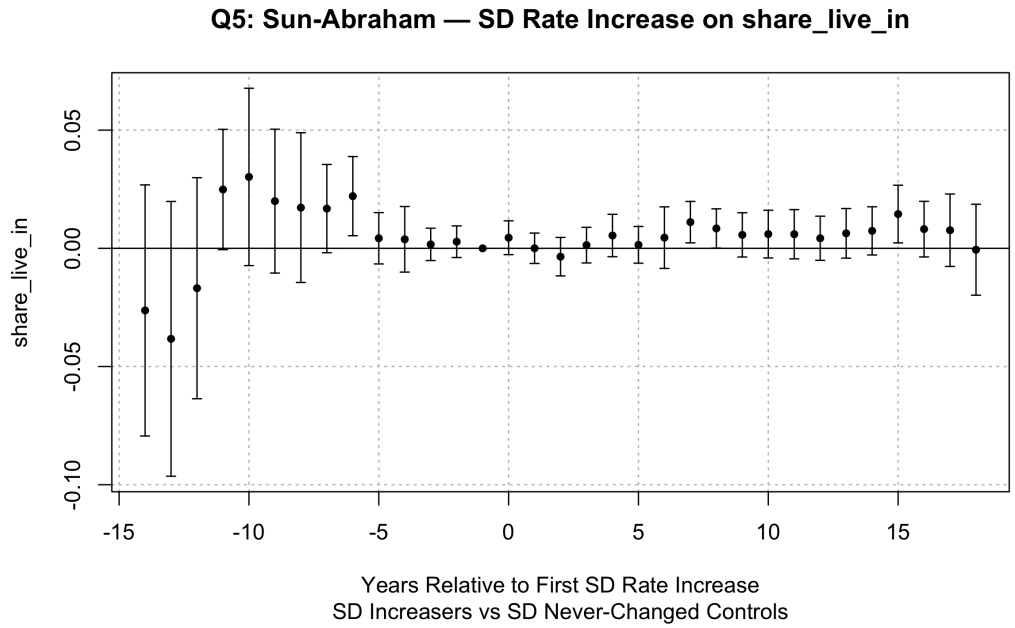


Figure 14: Q5: Sun-Abraham Event Study — SD Rate Increase on share_live_in. ATT = +0.005 (SE 0.004). Positive but not statistically significant on the share measure.

6.2.3 Q6: Is Sorting Concentrated at Municipal Boundaries?

Table 13: Q6: Within-City Tract-Level Effects — Boundary vs Interior

	(1)	(2)	(3)
	Within-City Share	Boundary Interaction	Total Workers × Boundary
Municipal EIT	0.0008$^{\cdot}$	0.0006*	0.0217***
	(0.0004)	(0.0003)	(0.0027)
Muni EIT × Boundary		0.0008	−0.0449$^{\cdot}$
		(0.0013)	(0.0266)
Outcome	share within city	share within city	log(total workers)
N	55,831	55,831	55,831
Boundary tracts		1,803 of 3,446 (52%)	
Fixed Effects		Tract, Year	
Clusters		Municipality	

Notes: Tract-level panel within PA municipalities. "Within-city share" = workers at tract j whose home is also in municipality i. Boundary = tract centroid within 1 km of nearest municipal boundary. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 14: Q6: Distance-to-Boundary Gradient

Distance bin	Muni EIT × distance bin
0–0.5 km	0.0022^{*} (0.0011)
0.5–1 km	0.0010 (0.0016)
1–2 km	0.0002 (0.0005)
2–5 km	0.0008^{**} (0.0003)
5+ km	-0.0437^{***} (0.0020)
N	55,831
Outcome	share within city

Notes: Distance from tract centroid to nearest municipal boundary. Tract and year FE. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

- The within-city share of workers is essentially unaffected by rate changes: $+0.0008^{\cdot}$ (SE 0.0004). This confirms that the city-level sorting result is not driven by within-city reshuffling—people are crossing municipal boundaries, not merely moving to a different neighborhood.

- The boundary interaction on the share outcome is null (+0.0008, SE 0.0013): boundary and interior tracts respond similarly. The sorting effect is not concentrated at municipal edges.

- However, boundary tracts show a marginally significant *decline in total workers* relative to interior tracts ($-0.045^{\cdot}$, SE 0.027). This suggests that boundary tracts may lose workers to adjacent municipalities, while interior tract employment remains stable.

- The distance gradient reveals a large negative effect at 5+ km (-0.044^{***}), likely reflecting composition: interior tracts in large cities (e.g., Philadelphia) respond differently from the typical small municipality tract. Tracts within 0–5 km of a boundary show small positive or null effects.

6.3 Heterogeneity

7 Robustness

Table 15: Robustness: Fiscal Controls and Philadelphia Exclusion

	TWFE			Dose-Response (treated only)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	+ Controls	No Philly	Baseline	+ Controls	No Philly
resident_eit	−0.0126*	−0.0104*	−0.0165**			
	(0.0052)	(0.0047)	(0.0060)			
rate × post				−0.0089**	−0.0079**	−0.0098**
				(0.0030)	(0.0028)	(0.0032)
N	27,277	14,174	27,257	3,058	2,173	3,038
Controls	No	Yes	No	No	Yes	No
Philadelphia	Yes	Yes	No	Yes	Yes	No

Notes: Controls include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid from PA DCED Annual Financial Reports. Philadelphia excluded in columns (3) and (6) due to its structurally different wage tax system. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Act 32 structural break across all outcomes. Table 16 reports the pre- vs post-Act 32 split across all key outcomes. The pattern is consistent: municipal EIT effects are concentrated in the post-2012 period when consolidated employer withholding made the tax binding. The school district decomposition reveals that the SD residential amenity effect (τ^{sd} on resident workers) is significant in both periods but *stronger* post-Act 32 ($+0.216^{***}$ vs $+0.101^{*}$), consistent with Act 32 making all EIT variation more salient.

Table 16: Act 32 Robustness: Pre/Post-2012 Split Across Outcomes

	(1) share_live_in	(2) log(Resident)	(3) log(Total Jobs)
Panel A: Municipal EIT only			
$\tau^m \times$ Pre-2012	-0.002 (0.004)	-0.010 (0.041)	-0.032 (0.033)
$\tau^m \times$ Post-2012	-0.018^{***} (0.005)	-0.198^{***} (0.057)	-0.127^{**} (0.043)
Panel B: Municipal + School District EIT			
$\tau^m \times$ Pre-2012	-0.002 (0.004)	-0.008 (0.041)	
$\tau^m \times$ Post-2012	-0.017^{***} (0.005)	-0.181^{**} (0.058)	
$\tau^{sd} \times$ Pre-2012	-0.001 (0.006)	0.101^{*} (0.048)	
$\tau^{sd} \times$ Post-2012	0.007 (0.007)	0.216^{***} (0.055)	
Panel C: Total EIT (municipal + school district)			
$\tau^{\text{total}} \times$ Pre-2012	-0.006 (0.004)		
$\tau^{\text{total}} \times$ Post-2012	-0.005 (0.004)		
N	27,273	27,273	27,273
Fixed Effects		Municipality, Year	
Clusters		Municipality	

Notes: Each cell is from a separate regression. Panel A: municipal EIT rate interacted with pre/post-2012 indicators. Panel B: both municipal and SD rates interacted with period indicators. Panel C: combined total EIT rate. The municipal effect is entirely post-2012; the SD amenity effect is present in both periods but stronger post-2012. Total EIT is null in both periods because the opposing municipal and SD effects cancel. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

- **Municipal EIT:** the pre-2012 coefficient is a precise zero across all outcomes. The post-2012 effect is significant for share_live_in (-0.018^{***}), resident workers (-0.198^{***}), and total jobs (-0.127^{**}). Enforcement is the mechanism.
- **School district EIT:** the resident worker amenity effect is positive and significant in both periods, but roughly doubles from $+0.101^{*}$ to $+0.216^{***}$ after Act 32. This suggests Act 32 made SD rate variation more salient alongside municipal rates.

- **Total EIT:** null in both periods (-0.006 and -0.005). The opposing municipal and SD components cancel regardless of the enforcement regime.

8 Discussion

9 Conclusion

Acknowledgments

References

[1] Tiebout, C. M. (1956). A Pure Theory of Local Expenditures. *Journal of Political Economy*, 64(5), 416–424.

[2] Sun, L. & Abraham, S. (2021). Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects. *Journal of Econometrics*, 225(2), 175–199.

[3] Callaway, B. & Sant'Anna, P. H. C. (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, 225(2), 200–230.

A Pre-Treatment Balance

Table 17: Pre-Treatment Balance: Treated vs. Never-Treated PA Municipalities

Variable	Treated Mean	Control Mean	Norm. Diff.	N (T / C)
Population	14,174	3,711	0.139	1,265 / 39,688
Total Revenue ($)	25.7M	2.9M	0.077	1,267 / 39,692
EIT Revenue ($)	4.4M	438K	0.062	
RE Tax Revenue ($)	2.9M	538K	0.129	
Police Expenditure ($)	3.1M	424K	0.110	
Streets Expenditure ($)	1.5M	496K	0.121	
EIT Rate (%)	0.50	0.50	0.055	

Notes: Treated = municipalities that raised EIT during the sample period (pre-treatment years only). Control = municipalities that never raised EIT. All normalized differences < 0.15, within standard balance thresholds. Source: PA DCED Annual Financial Reports.

B Pre-Trend Tests on Fiscal Observables

We test the parallel trends assumption on six fiscal observables from the PA DCED Annual Financial Reports using the Sun and Abraham (2021) estimator. For each outcome, we estimate Equation 2 with municipality and year fixed effects, clustering standard errors at the municipality level. Table 18 reports the maximum absolute t-statistic among the pre-treatment coefficients at relative times $\ell \in \{-5, -4, -3, -2\}$.

Pre-trends are flat for five of six fiscal observables: total revenue (max $|t| = 1.60$), EIT revenue (1.33), streets expenditure (0.80), population (1.09), and total debt (1.36). None of the pre-treatment coefficients for these outcomes are individually significant at the 5% level.

Police expenditure is the exception. The pre-treatment coefficients at $\ell = -5$ and $\ell = -4$ are negative and significant ($t = -2.85$ and $t = -2.42$, respectively), indicating that treated municipalities had relatively lower police spending 4–5 years before the rate increase.

This pattern suggests that some EIT increases may have been associated with anticipated public-safety expansions financed by the new revenue. Importantly, the main results on `share_live_in` are robust to controlling for log police expenditure (Table 15), confirming that this pre-trend does not confound the estimated sorting response.

Table 18: Pre-Trend Tests: Sun-Abraham on Fiscal Observables (PA, 2002–2019)

	Pre-Treatment Coefficients					
Outcome	$\ell = -5$	$\ell = -4$	$\ell = -3$	$\ell = -2$	Max $\lvert t \rvert$	N
log(Total Revenue)	−0.076	−0.051	−0.012	−0.052	1.60	43,170
	(0.049)	(0.056)	(0.038)	(0.032)		
log(EIT Revenue)	−0.025	−0.033	0.027	0.049	1.33	42,278
	(0.051)	(0.052)	(0.032)	(0.037)		
log(Police Exp)	-0.085^{***}	-0.068^{**}	0.011	0.037	2.85	23,464
	(0.030)	(0.028)	(0.052)	(0.039)		
log(Streets Exp)	−0.001	0.003	−0.043	−0.029	0.80	43,105
	(0.059)	(0.075)	(0.053)	(0.059)		
log(Population)	0.011	0.007	0.002	0.003	1.09	43,150
	(0.012)	(0.008)	(0.006)	(0.003)		
log(Total Debt)	−0.029	−0.051	−0.179	−0.103	1.36	24,863
	(0.176)	(0.214)	(0.132)	(0.102)		

Notes: Each row reports Sun and Abraham (2021) estimates of the pre-treatment coefficients $\hat{\mu}_\ell$ for $\ell \in \{-5, -4, -3, -2\}$, with $\ell = -1$ as the reference period. Standard errors (in parentheses) are clustered at the municipality level. Max $|t|$ is the largest absolute t-statistic among the four pre-treatment coefficients. Pre-trends are flat for all outcomes except police expenditure, where treated municipalities had significantly lower spending at $\ell = -5$ and $\ell = -4$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PA DCED Annual Financial Reports.

C Additional Figures